TABLEAU

Module 5 - Creating Groups, Sets & Hierarchies

What You Will Learn in This Session

Topic 1	Groups — How to bundle values into custom categories with your own labels
Topic 2	Sets — How to create smart IN/OUT segments based on conditions or manual picks
Topic 3	Hierarchies — How to build drill-down structures like Category → Product → Variant
Bonus	Groups vs Sets vs Hierarchies — the complete comparison table

Beginner-Friendly • Real-Life Examples • Same Dataset Throughout

Our Practice Dataset — Module 5

We use the same retail shop dataset from Module 4, now extended with a Region column and City details. This makes Groups, Sets, and Hierarchies more meaningful and realistic.

Row	Region	City	Category	Product	Sales (₹)	Profit (₹)	Qty	Month
1	West	Mumbai	Electronics	Laptop	62,000	14,000	1	January
2	North	Delhi	Electronics	Laptop	55,000	11,000	1	January
3	West	Mumbai	Electronics	Mobile Phone	36,000	8,000	2	January
4	South	Bangalore	Electronics	Mobile Phone	32,000	7,200	2	February
5	South	Chennai	Electronics	Tablet	25,000	5,500	1	February
6	North	Delhi	Electronics	Tablet	22,000	4,800	1	February
7	West	Mumbai	Accessories	Headphones	10,500	3,200	3	March
8	West	Pune	Accessories	Headphones	8,400	2,600	2	March
9	North	Delhi	Accessories	Keyboard	6,200	1,800	2	March
10	South	Chennai	Accessories	Keyboard	4,800	1,200	3	March

What's New in This Dataset?

We added a 'Region' column: West (Mumbai, Pune), North (Delhi), South (Bangalore, Chennai). This gives us a natural hierarchy: Region → City → Category → Product. We will use this throughout all three topics.

Topic 1 — Groups in Tableau

A Group lets you combine multiple values of a field into a single, custom bucket with your own label. Instead of seeing 5 individual cities, you might want to see just 3 zones — West, North, South. That is exactly what a Group does.

Real-life analogy

Think of how the Indian Premier League organises cricket teams. Instead of showing all 10 teams individually every time, broadcasters group them: 'Top 4 Playoff Teams' vs 'Remaining Teams'. They have not changed the teams — they have just created a grouping that makes the story clearer. That is exactly what a Tableau Group does.

1.1 What Problem Does a Group Solve?

Imagine your dataset has 28 Indian states as individual values. Your chart becomes unreadable. But if you group them into North India, South India, East India, West India — suddenly the chart is clean, clear, and tells a story.

In our dataset, we have 5 cities. We want to group them into 3 zones:

City	Original (No Group)	After Grouping into Zones
Mumbai	Mumbai	West Zone
Pune	Pune	West Zone
Delhi	Delhi	North Zone
Bangalore	Bangalore	South Zone
Chennai	Chennai	South Zone

After grouping, your chart shows 3 bars (West Zone, North Zone, South Zone) instead of 5 bars (all individual cities). Much easier to present to a manager!

1.2 How to Create a Group — Method 1: Select on Chart

The fastest and most visual way — you select data points directly on your chart and group them instantly.

1	Build a bar chart: City on Rows, SUM(Sales) on Columns. You should see 5 bars — one for each city
2	On the chart, click the bar for 'Mumbai'. Then hold Ctrl (Windows) or Cmd (Mac) and click 'Pune' as well — both bars are now selected and highlighted

3	A small tooltip/context menu appears near your selection — click the paperclip icon (Group Members) or right-click → Group
4	Tableau instantly creates a new group combining Mumbai and Pune into one item. A new field appears in the Data panel called 'City (group)'
5	The group is named 'Mumbai, Pune' by default — double-click that label on the chart or in the Data panel to rename it to 'West Zone'
6	Repeat for Delhi (click Delhi → right-click → Group) — rename to 'North Zone'
7	Select both Bangalore and Chennai together (Ctrl+click both) → Group → rename to 'South Zone'
8	Your chart now shows 3 clean zone bars instead of 5 city bars

1.3 How to Create a Group — Method 2: From Data Panel

A more controlled method — useful when you are working with many values and want to set up groups before building the chart.

1	In the Data panel on the left, find the 'City' field
2	Right-click 'City' → Create → Group
3	A 'Create Group' dialog opens. You see all 5 cities listed on the left side
4	Click 'Mumbai', then hold Ctrl and click 'Pune' to select both
5	Click the 'Group' button in the dialog — they merge into one group
6	Double-click the group name at the top right and rename it 'West Zone'
7	Select 'Delhi' alone → click Group → rename to 'North Zone'
8	Select 'Bangalore' and 'Chennai' together → Group → rename to 'South Zone'
9	Click OK — a new field 'City (group)' appears at the bottom of the Dimensions section in the Data panel with a small paperclip icon

1.4 The 'Other' Group — What It Is and When to Use It

What is 'Other'?

When you create groups, Tableau gives you an option called 'Include Other'. Any value that you did NOT manually assign to a group gets placed into a bucket called 'Other'. This is very useful when you have 50 cities but only want to highlight the top 5 — everything else becomes 'Other'.

Practical example — Group only the top 2 cities, rest become 'Other':

City	Sales (₹)	Group Assignment
Mumbai	1,08,500	Star Cities (manually grouped)
Delhi	83,200	Star Cities (manually grouped)
Bangalore	32,000	Other (auto-assigned)
Chennai	29,800	Other (auto-assigned)
Pune	8,400	Other (auto-assigned)

How to enable 'Other':

- In the 'Create Group' dialog, check the box at the bottom: 'Include Other'
- Any value you do not group will automatically fall into the 'Other' category
- You can rename 'Other' to something more meaningful — e.g. 'Rest of India'

1.5 Editing an Existing Group

1	In the Data panel, right-click the grouped field (e.g. 'City (group)') — it has a paperclip icon
2	Click 'Edit Group'
3	The same Create Group dialog opens, showing your existing groups
4	To add a new city to a group: click the city in the left list, then click the group name on the right → click 'Add to'
5	To rename a group: double-click its name on the right side
6	To ungroup: select the group → click 'Ungroup'
7	Click OK to save changes

1.6 Practical Exercise — Product Groups

Let us group our 5 products into 2 business segments: Premium (high-value items) and Budget (low-cost accessories).

Product	Sales (₹) per unit	Group Name	Business Logic
Laptop	₹55,000–62,000	Premium	High-ticket electronics
Mobile Phone	₹16,000–18,000	Premium	Mid-to-high range device
Tablet	₹22,000–25,000	Premium	High-value electronics
Headphones	₹3,500–4,200	Budget	Low-cost accessory
Keyboard	₹2,400–3,100	Budget	Low-cost accessory

After creating this group:

- A new field appears: **'Product (group)'** with only 2 values: Premium, Budget
- Drag 'Product (group)' to Rows and SUM(Sales) to Columns
- Instantly see total revenue from Premium (₹2,32,000) vs Budget (₹28,600)
- This tells a clear story: 89% of revenue comes from Premium products

Key Thing to Remember About Groups

A Group is STATIC — you manually decide who belongs where. It does not update automatically if new data comes in. If you add a new city 'Hyderabad' to your data tomorrow, it will be placed in 'Other' and you must manually add it to a zone. For dynamic categorisation, use a Calculated Field (IF/CASE) or a Set instead.

Topic 2 — Sets in Tableau

A Set is a custom field that divides your data into exactly two groups: IN (members that belong to the set) and OUT (everyone else). Sets are more powerful than Groups because they can be dynamic — automatically updating based on conditions as your data changes.

Real-life analogy

Think of a club membership. A gym has members and non-members. Members are IN the set — they get access to the gym. Non-members are OUT. Every month, new people join and some quit — the IN/OUT membership updates automatically. A Dynamic Set in Tableau works the same way — it checks conditions and automatically decides who is IN and who is OUT.

2.1 Group vs Set — The Key Difference

Group	Set
Creates multiple buckets with custom names	Creates exactly TWO states: IN and OUT
Static — you manually assign each value	Can be Dynamic — updates with data automatically
Shows the group name in the chart	Shows IN/OUT or TRUE/FALSE in the chart
Best for: renaming and reorganising categories	Best for: segmenting, highlighting, comparing
Example: City → West Zone, North Zone, South Zone	Example: High Sales Cities (IN) vs rest (OUT)
Cannot be combined with other groups easily	Multiple sets can be combined (Combined Sets)

2.2 Type 1 — Static Set (Manual Selection)

Simple Definition

A Static Set is one where YOU manually choose which values belong in the set. The membership never changes unless you manually edit it. Like a VIP guest list that you personally curate.

Real-life example: *A company decides its 'Key Accounts' are the 3 specific clients that have been with them for 10+ years. This list is fixed — it does not change based on sales. They manually pick those 3 clients for the set.*

Example — Create a Static Set: 'Focus Cities' = Mumbai and Delhi only

1	In the Data panel, right-click the 'City' field
2	Click Create → Set
3	The 'Create Set' dialog opens. You are on the 'General' tab by default
4	You see a list of all 5 cities. Check the boxes for 'Mumbai' and 'Delhi' only
5	Name your Set at the top: type 'Focus Cities'
6	Click OK — a new field 'Focus Cities' appears at the BOTTOM of the Data panel in a separate 'Sets' section, with a Venn diagram icon
7	This set has exactly 2 values: IN (Mumbai, Delhi) and OUT (Bangalore, Chennai, Pune)

2.3 Type 2 — Dynamic Set (Condition-Based)

Simple Definition

A Dynamic Set automatically decides who is IN based on a rule or condition you set — like 'all cities where total Sales > ₹50,000'. As your data changes and a city crosses that threshold, it automatically joins the set. No manual updating needed.

Real-life example: *A bank automatically classifies customers as 'Premium' if their account balance exceeds ₹10 lakhs. As customers save more money and cross the threshold, they automatically get added to the Premium set. The bank does not manually update the list every day — the rule does it.*

Example — Create a Dynamic Set: 'High Sales Cities' = cities where SUM(Sales) > ₹50,000

1	Right-click 'City' in the Data panel → Create → Set
2	In the 'Create Set' dialog, click the 'Condition' tab (second tab at the top)
3	Select 'By Field' radio button
4	Set the condition: Sales → Sum → >= → 50000
5	Name your set: 'High Sales Cities'
6	Click OK — Tableau evaluates which cities meet this condition

Which cities are IN and OUT?

City	Total Sales (₹)	Condition (>= ₹50,000)	Set Membership
Mumbai	1,08,500	✓ 1,08,500 ≥ 50,000	IN ✓
Delhi	83,200	✓ 83,200 ≥ 50,000	IN ✓
Bangalore	32,000	✗ 32,000 < 50,000	OUT ✗
Chennai	29,800	✗ 29,800 < 50,000	OUT ✗
Pune	8,400	✗ 8,400 < 50,000	OUT ✗

Mumbai and Delhi are IN the set. The other 3 cities are OUT. If next month Bangalore's sales jump to ₹60,000, it will automatically move to IN — you do not have to do anything!

2.4 Type 3 — Top/Bottom N Set

A special type of Dynamic Set where membership is based on ranking — automatically the Top N or Bottom N members of a field.

1	Right-click 'Product' in the Data panel → Create → Set
2	Click the 'Top' tab in the Create Set dialog
3	Select 'By Field' → Top → 3 → Sales → Sum
4	Name it 'Top 3 Products'
5	Click OK
6	IN: Laptop, Mobile Phone, Tablet (top 3 by sales) OUT: Headphones, Keyboard (bottom 2)

Powerful Use of Sets on Dashboards

Place your Set as a filter on the dashboard. Add a parameter or action that lets users click a city on one chart to toggle it between IN and OUT. This creates a dynamic 'selection highlight' that updates all charts simultaneously — this is called a Set Action and is used in advanced dashboards.

Topic 3 — Hierarchies in Tableau

A Hierarchy is a structured drill-down path through your data — from a big general level to smaller specific details. It lets you click a + button on your chart to zoom in from Region → City → Category → Product, and click − to zoom back out.

Real-life analogy

Think of Google Maps. You start zoomed out seeing the whole of India. You click on Maharashtra — it zooms in. You click Mumbai — zooms in further to neighbourhoods. You click Bandra — zooms to streets. Each zoom level is a level in a hierarchy: Country → State → City → Area → Street. Tableau Hierarchies work exactly the same way for your data.

3.1 Why Hierarchies Are So Useful

- They save space — start with a summary, drill into details only when needed
- They make presentations interactive — managers can explore at their own pace
- They replace the need for multiple separate charts — one chart serves all levels
- They match how people naturally think: from big picture to specific details

In our dataset, we have a natural hierarchy:

Our Data Hierarchy
Region (Level 1 — broadest)
↳ City (Level 2)
↳ Category (Level 3)
↳ Product (Level 4 — most detailed)

With this hierarchy, you can start a chart at Region level, click + on 'West' to see cities within West, click + on 'Mumbai' to see categories within Mumbai, and click + again to see products.

3.2 The Built-in Date Hierarchy

Tableau automatically creates a Date Hierarchy for any date field you connect. You do not have to build it — it is already there!

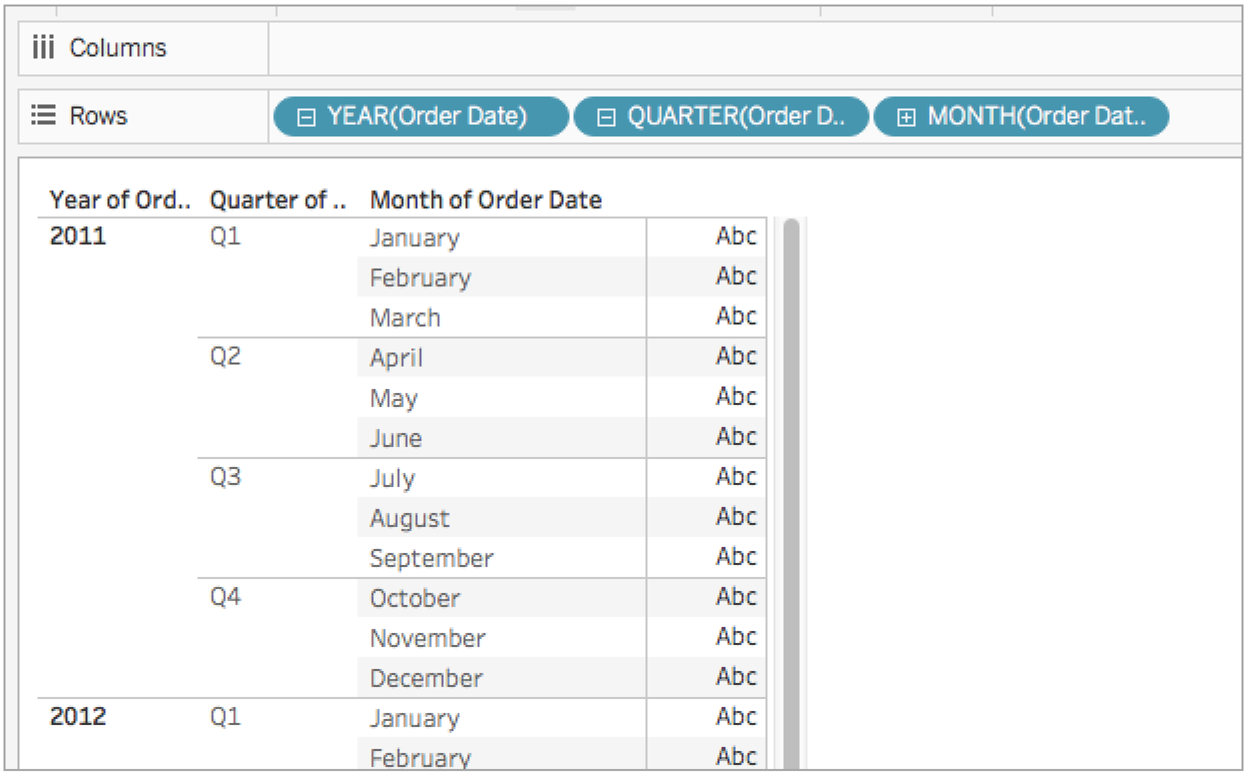
Tableau's Automatic Date Hierarchy
Year (e.g. 2024)
↳ Quarter (e.g. Q1, Q2, Q3, Q4)

↳ Month (e.g. January, February)

↳ Week (e.g. Week 1, Week 2)

↳ Day (e.g. 01, 02, 03…)

Real-life example: *Your sales dashboard starts showing Total Sales for 2024 as one bar. You click the + next to 2024 — it splits into Q1, Q2, Q3, Q4. Click + on Q1 — it splits into January, February, March. This is the date hierarchy working automatically.*



How to use the Date Hierarchy
Drag your Order Date field to the Columns shelf. Tableau defaults to Year. Click the + sign that appears on the Year header in your chart. It drills down to Quarter. Click + again for Month. Click the − sign to drill back up. You are using the hierarchy automatically — no setup needed.

3.3 Creating a Custom Hierarchy — Step by Step

For non-date fields, you must build the hierarchy yourself. It takes about 30 seconds and makes your charts dramatically more interactive.

We will build: Region → City → Category → Product

1	In the Data panel, find the 'Product' field (this will be the LOWEST/innermost level)
2	Drag 'Product' and DROP IT directly ON TOP of 'Category' in the Data panel A small hierarchy icon appears — Tableau asks if you want to create a hierarchy
3	A dialog appears: 'Create Hierarchy'. Type a name: 'Product Hierarchy' → Click OK
4	You will now see 'Product Hierarchy' in the Data panel with Category and Product nested inside it (with indent)
5	Now drag 'City' and drop it ON TOP of 'Category' inside the hierarchy — it adds City as a level above Category
6	Drag 'Region' and drop it ON TOP of 'City' — Region becomes the outermost (highest) level
7	Your hierarchy now shows: Product Hierarchy → Region → City → Category → Product (in that order)
8	If the order is wrong: inside the hierarchy, drag levels up or down to reorder them

What your completed hierarchy looks like in the Data panel:

Product Hierarchy (in Data Panel)
Product Hierarchy
↳ Region
↳ City
↳ Category
↳ Product

3.4 Using the Hierarchy in a Chart — Drill Down and Up

Once your hierarchy is built, using it is very intuitive. The chart handles the drill-down for you automatically.

1	Drag 'Region' (from inside Product Hierarchy) to the Rows shelf — OR drag the entire hierarchy to Rows
2	Drag SUM(Sales) to Columns — you see 3 bars: North, South, West

3	Look at the Region labels on the chart — you see a small '+' sign next to each region name
4	Click the '+' next to 'West' — it EXPANDS to show Cities within West: Mumbai, Pune
5	Click '+' next to 'Mumbai' — it expands to show Categories: Accessories, Electronics
6	Click '+' next to 'Electronics' — it shows Products: Laptop, Mobile Phone, Tablet
7	Click the '−' sign next to any level to COLLAPSE back up to the parent level
8	To expand ALL members at once: right-click any '+' icon → 'Expand All'

What you see at each drill-down level:

Level	What the Chart Shows	Bar Count	Total Sales (₹)
Region (Level 1)	North, South, West	3 bars	3 region totals
Drill into West	Mumbai, Pune (within West)	2 bars	1,08,500 + 8,400
Drill into Mumbai	Accessories, Electronics (in Mumbai)	2 bars	10,500 + 98,000
Drill into Electronics	Laptop, Mobile Phone (in Mumbai, Electronics)	2 bars	62,000 + 36,000

3.5 Editing and Removing a Hierarchy

- **To rename:** Right-click the hierarchy name in the Data panel → Rename
- **To add a level:** Drag any field from the Data panel and drop it inside the hierarchy at the right position
- **To remove a level:** Right-click the field INSIDE the hierarchy → Remove from Hierarchy (the field goes back to the main Data panel)
- **To delete the whole hierarchy:** Right-click the hierarchy name → Remove Hierarchy (all fields go back to normal)

3.6 Practical Exercise — Date Hierarchy on Sales Data

Using the Order Date field with the built-in date hierarchy, let us see how sales trend from year down to month.

1	Drag 'Month' (or 'Order Date') to the Columns shelf
2	Drag SUM(Sales) to Rows — you get a line chart

3	Tableau shows data by Year by default. Right-click the date field on Columns → choose 'Month' from the second group (Continuous) to see months
4	To use hierarchy drill-down: put YEAR(Order Date) on Columns — you see one point for 2024
5	Click the '+' on 2024 — it splits into Q1, Q2, Q3 (our data has Jan-Mar only)
6	Click '+' on Q1 — it shows January, February, March as separate points
7	You have drilled from Annual → Quarterly → Monthly without changing the chart setup

Tip — Date Hierarchy vs Date Parts

Using the hierarchy drill-down (Year → Quarter → Month) keeps all levels connected and lets you go up and down. Using DATEPART() in a calculated field or right-clicking to choose a date level gives you a fixed view at one level only. For interactive exploration, always use the hierarchy. For a fixed monthly report, use the date part directly.

The Big Comparison — Groups vs Sets vs Hierarchies

These three features all organise your data differently. Knowing WHEN to use which one is the key skill.

Criteria	Groups	Sets	Hierarchies
Purpose	Rename & bundle multiple values into custom categories	Divide data into IN (member) vs OUT (non-member)	Create drill-down levels from broad to specific
Output	Multiple named buckets (West Zone, North Zone)	Binary: IN or OUT	Nested levels with + / − expand controls
Static or Dynamic	Static only — manual assignment	Both Static (manual) and Dynamic (condition-based)	Static — you define the levels once
Number of buckets	As many as you need	Always exactly 2: IN and OUT	As many levels as you add
Updates automatically?	No — must edit manually	Yes (if Dynamic) — updates as data changes	No — structure is fixed
Use on Color?	Yes — each group gets a colour	Yes — IN is one colour, OUT is another	Not directly
Use as Filter?	Yes	Yes — filter to show only IN members	Yes — filter at any level
Best for	Regrouping messy categories	Segmenting and highlighting key performers	Drill-down exploration in dashboards
Icon in Data panel	Paperclip icon on field	Venn diagram icon — separate Sets section	Hierarchy icon with nested fields
Real-life example	Grouping states into North/South/East/West	VIP customers vs Regular customers	Country → State → City → Pincode

Module 5 — Quick Reference Cheat Sheet

Groups — Quick Steps

```
METHOD 1 (On Chart):  Select bars/marks with Ctrl+Click → right-click →
Group → rename

METHOD 2 (Data panel): Right-click field → Create → Group → select values →
Group → rename

EDIT:   Right-click grouped field (paperclip icon) → Edit Group

OTHER:  Check 'Include Other' in dialog to catch unassigned values
automatically

REMEMBER: Groups are STATIC — manually assign + manually update when new
data arrives
```

Sets — Quick Steps

```
STATIC SET:    Right-click field → Create → Set → General tab → check values
manually

DYNAMIC SET:   Right-click field → Create → Set → Condition tab → By Field →
set rule

TOP N SET:     Right-click field → Create → Set → Top tab → Top N → by
measure

COMBINED SET:  Right-click any set → Create Combined Set → choose 2 sets +
combination type

USE ON CHART:  Drag Set to Color shelf (IN vs OUT colours) or Filters shelf
(show IN only)

REMEMBER: Dynamic Sets update automatically — no manual editing needed when
data changes
```

Hierarchies — Quick Steps

```
CREATE:   Drag the CHILD field and drop it ON TOP of the PARENT field in
Data panel

ADD LEVEL: Drag another field and drop it inside the hierarchy at the right
position

USE:      Drag top-level field to Rows/Columns → click + to drill down, – to
drill up

DATE:     Date hierarchy is AUTOMATIC — just drag a date field and click +
on Year

REMOVE:   Right-click field inside hierarchy → Remove from Hierarchy

REMEMBER: Click + on chart headers to drill DOWN, click – to drill UP
```

When to Use Which — Decision Guide

Your Goal	Use This
Rename 5 cities into 3 zones with custom labels	Group
Highlight top performers in orange vs rest in grey	Set (drag to Color)
Let users click to drill from Region → City → Product	Hierarchy
Auto-segment customers: spending > ₹1 lakh = Premium	Dynamic Set (Condition)
Manually pick your VIP clients for special reporting	Static Set (General tab)
Group shirt sizes S, M, L together as 'Small Sizes'	Group
Find cities in BOTH Top Sales AND Top Profit	Combined Set (intersection)
See Year → Quarter → Month trend by clicking the chart	Date Hierarchy
Create a 'rest of India' bucket for ungrouped states	Group with 'Include Other'

End of Module 5 — Groups, Sets & Hierarchies in Tableau

Practice: Group the 5 cities into 3 zones, create a Dynamic Set for High Sales Products, and build a Region → City → Product hierarchy on the dataset!